1141 CS140 資訊概論　小考 1 試題　2025/09/26　90:10-10:00am

學號：____________________　**姓名：**________________________　**作答後交回**

題號	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
得分	13	10	6	6	6	5	5	8	12	5	4	4	4	4	4	4

1. (17分) Computers built after 1950 more or less follow the von Neumann model. They have become faster, smaller, and cheaper, but the principle is almost the same. Historians divide this period into generations, with each generation witnessing some major change in hardware and software.

(A) The von Neumann Model defines the components of a computer, which are (每格2分)

(a) memory　(b) arithmetic logic unit (ALU)　(c) control unit (CU)　(d) input/output (I/O)

(B) Match the computer generations with their corresponding descriptions. (連線配對，每項1分)

First Generation (1950–1959)	Marked by the emergence of commercial computers, which were bulky, used vacuum tubes as electronic switches, and were operated only by professionals.
Second Generation (1959–1965)	Characterized by the replacement of vacuum tubes with transistors, resulting in faster and more reliable machines.
Third Generation (1965–1975)	Notable for the introduction of microcomputers, making computing more accessible.
Fourth Generation (1975–1985)	Defined by the invention of the integrated circuit, which further reduced the cost and size of computers while increasing efficiency.
Fifth Generation (1985–present)	Distinguished by the development of laptops and palmtops, advances in secondary storage media, the rise of multimedia applications and virtual reality.

2. Convert each number to the target base. 不同基底數字轉換 (每格2分)

(a) $(56)_{16}$ = (01010110　　　　$)_2$

(b) $(125.125)_{10}$ = (1111101.001　　　　$)_2$

(c) $(1111.11)_2$ = (15.75　　$)_{10}$ = (17.6　　$)_8$ = (F.C　　$)_{16}$

3. Convert the following decimal numbers to **sign-and-magnitude** representations with 8-bit allocation.

(a) $(127)_{10}$ = (　01111111　　$)_2$　　(每格 3 分)

(b) $(–127)_{10}$ = (　11111111　　$)_2$

4. (6 分) Convert the following decimal numbers to their 8-bit **two's complement** representations.

(a) $(127)_{10}$ = (　01111111　　$)_2$　　(每格 3 分)

(b) $(–127)_{10}$ = (　10000001　　$)_2$

5. What is the maximum integer number and minimum integer number that can be represented in two's complement numbers with 10-bit allocation? (每格 3 分)

(a) Maximum integer：511

(b) Minimum integer　：–512

6. What is the equivalent decimal value of the following 32-bit IEEE 754 floating point representation? (1-bit sign, 8-bit exponent using Excess_127 representation, 23-bit mantissa) (5 分)

   11000000011100000000000000000000

   Ans: −3.75

7. Write the 32-bit IEEE 754 floating point representation for the number 168.875 (5 分)

   Ans: 01000011001010001110000000000000

8. Decide whether the operation creates an overflow if the numbers and the result are represented in 8-bit two's complement representation? (Fill in Yes or No) (每格 2 分)

   (a)__Yes__00000101 + 01111101 (b) __No__ 11000010 + 11111111

   (c)__No__11000010 + 00111111 (d)__No__ 00000010 + 11111111

9. Given the bit patterns p = 10011001 and q = 01100110, both of them are represented in two's complement format, write the result after (每格 3 分，答案寫二進位數值)

   (a) performing p **OR** q:__11111111__

   (b) performing p **XOR** q:__11111111__

   (c) performing an arithmetic left shift operation on p: __00110010__

   (b) performing a logic left shift operation on p: __00110010__

10 An audio signal is sampled 8000 times per second. Each sample is represented by 256 different level. How many bits per second are needed to represent this signal?

   Ans:__(8000 sample/ sec) × (8 bits / sample) = 64,000 bits /seconds__

以下為選擇題，每題 4 分

11 One of the first computers based on the von Neumann model was called ________.

   __a__ (a) EDVAC (b) ABC (c) COBAL (d) Pascaline (e) Jacquard loom

12 A 32-bit code called ________ represents symbols in all languages.

   __d__(a) ANSI (b) ASCII (c) Extended ASCII (d) Unicode (e) EBCDIC

13 The equivalent of one's complement in the binary system is nine's complement in the decimal system (1 = 2 − 1 and 9 = 10 − 1). With n-digit allocation, we can represent nine's complement numbers in the range of $-[(10^n/2) - 1]$ to $+[(10^n/2) - 1]$. The nine's complement of a number with n digit allocation is obtained as follows. If the number is positive, the nine's complement of the number is itself. If the number is negative, we subtract each digit from 9. In three-digit allocation, what is the range of the numbers we can represent using nine's complement?

   __d__ (a)−500 to 500 (b) −500 to 499 (c) −499 to 500 (d) −499 to 499 (e) none of above

14 Assuming three-digit allocation, the nine's complement of the decimal number −234 is ____.

   __c__ (a)233 (b) 234 (c) 765 (d) 766 (e) none of above

15 The equivalent of two's complement in the binary system is ten's complement in the decimal system (in the binary system, 2 is the base, in the decimal system, 10 is the base). Using n-digit allocation, we can represent ten's complement numbers in the range of $-(10^n/2)$ to $+[(10^n/2) - 1]$. The ten's complement of a number with n digit allocation is obtained by first find the nine's complement of the number and then adding 1 to the result. In three-digit allocation, what is the range of the numbers we can represent using ten's complement?

   __b__ (a)−500 to 500 (b) −500 to 499 (c) −499 to 500 (d) −499 to 499 (e) none of above

16 Assuming three-digit allocation, the ten's complement of the decimal number −234 is ____.

   __d__ (a) 233 (b) 234 (c) 765 (d) 766 (e) none of above